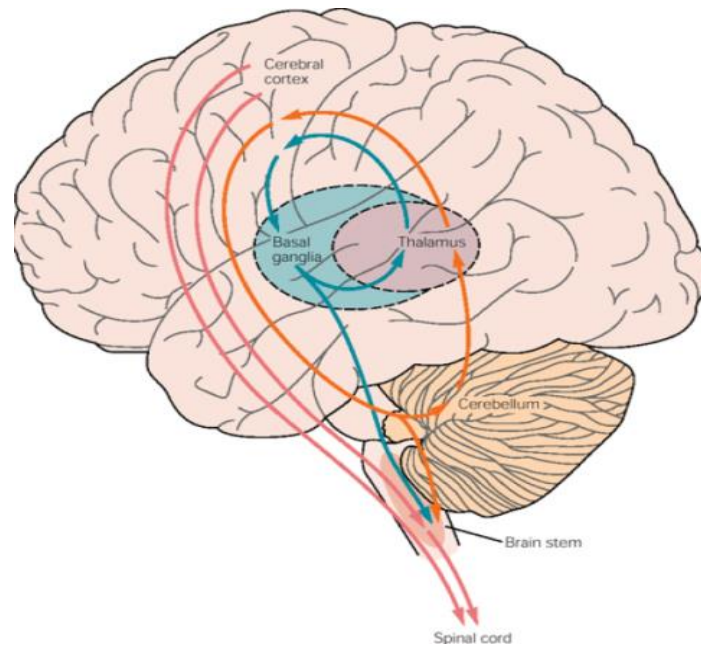


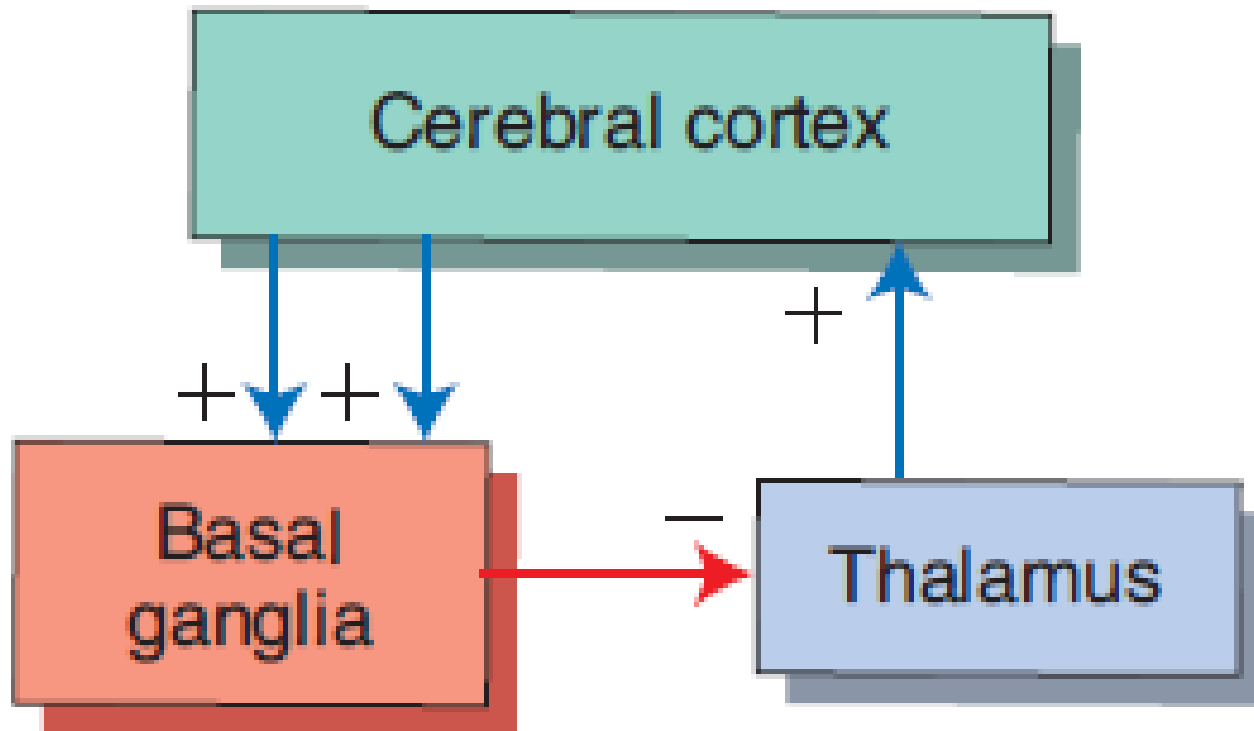
Nervous System



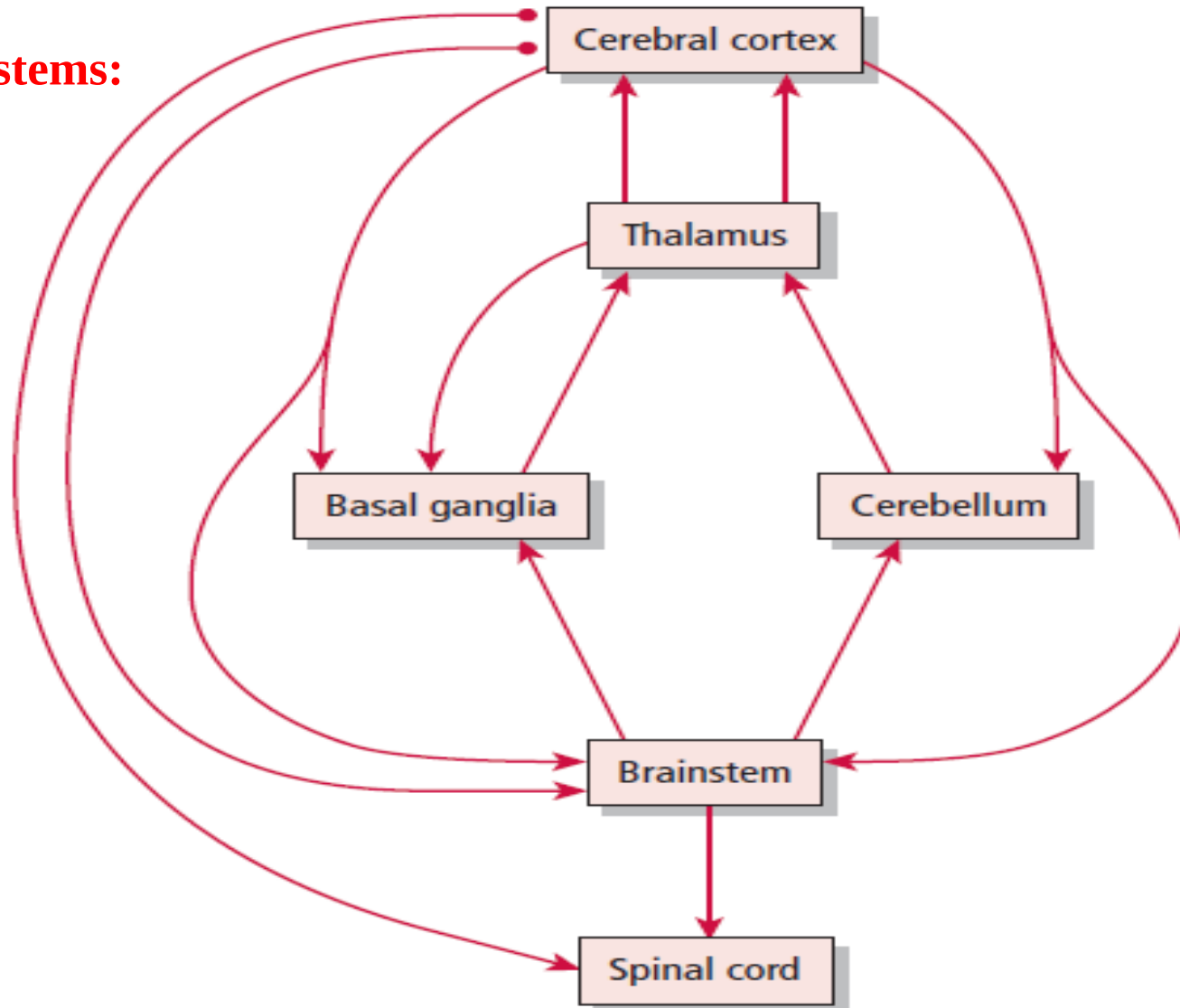
Basal Ganglia

The Basal Ganglia

- Deep nuclei of the cerebrum
- Aid the motor cortex in planning + generating motor actions (*Normal voluntary movement*):
 - Skeletomotor + oculomotor...
- Cognitive + affective behavior...
- **NB**
 - No direct input or output connections with the spinal cord



The link between cerebral cortex and basal ganglia

Motor Systems:

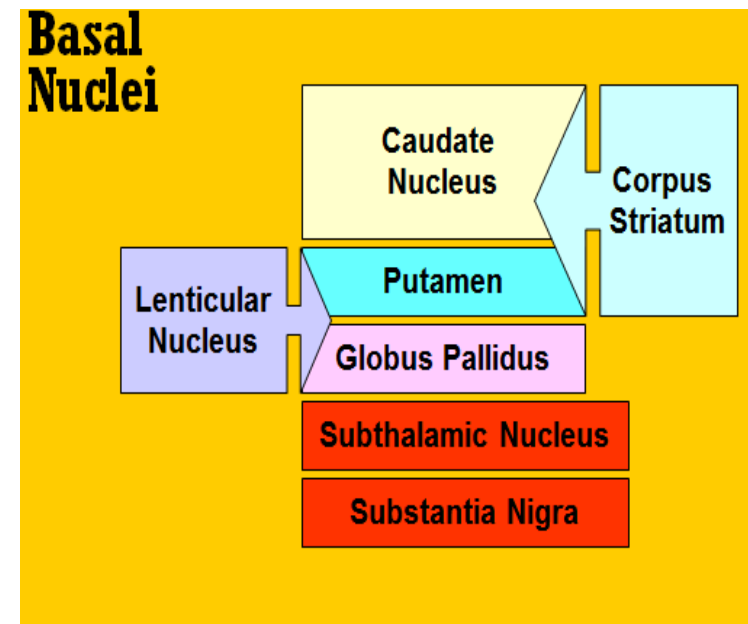
Connections between the cerebral cortex, thalamus, basal ganglia, cerebellum, brainstem, & spinal cord.

Physiologic Anatomy

- Three large nuclear masses:
 - Caudate nucleus
 - Putamen
 - Globus pallidus/**GPe + GPi**

Striatum/Neostriatum

Lentiform/lenticular nucleus
- Two functionally related nuclei:
 - Subthalamic nucleus (**STN**)
 - Substantia nigra (**SNpc + SNpr**)

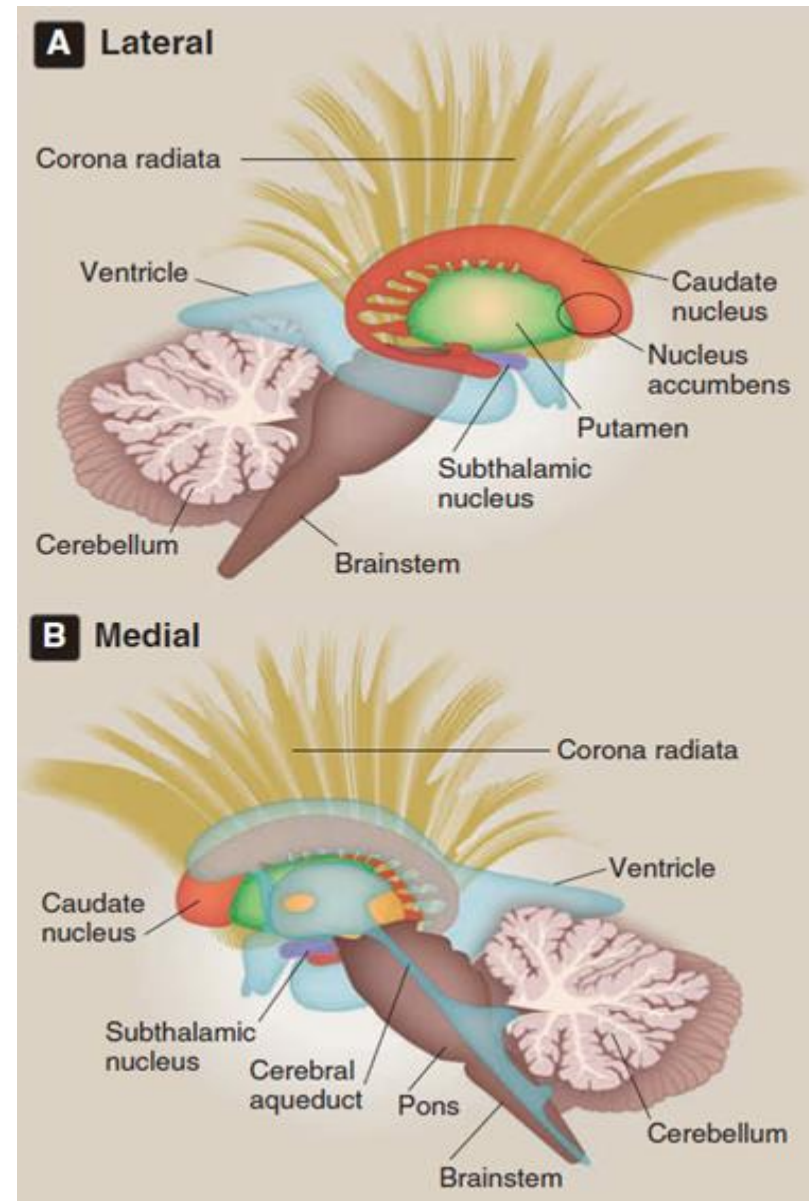


Organization of the Basal Ganglia

- Input nuclei/zones
 - Caudate nucleus & Putamen (ACh, GABA)
 - Nucleus accumbens (DA)
- Intrinsic nuclei
 - Globus pallidus (GPe, GABA)
 - Subthalamic nucleus (Glutamate)
 - Substantia nigra (SNpc, DA)
- Output nuclei
 - Globus pallidus (GPi) & Substantia nigra (SNpr) (GABA)

Caudate Nucleus

- A tadpole-shaped nucleus
- Its head lies in the floor of the lateral ventricle + its body arches over the thalamus in a **C shape** tapering off into a tail lying in the roof of the inferior horn of the lateral ventricle
- **Putamen + caudate nuclei** are the **input nuclei** to the basal ganglia

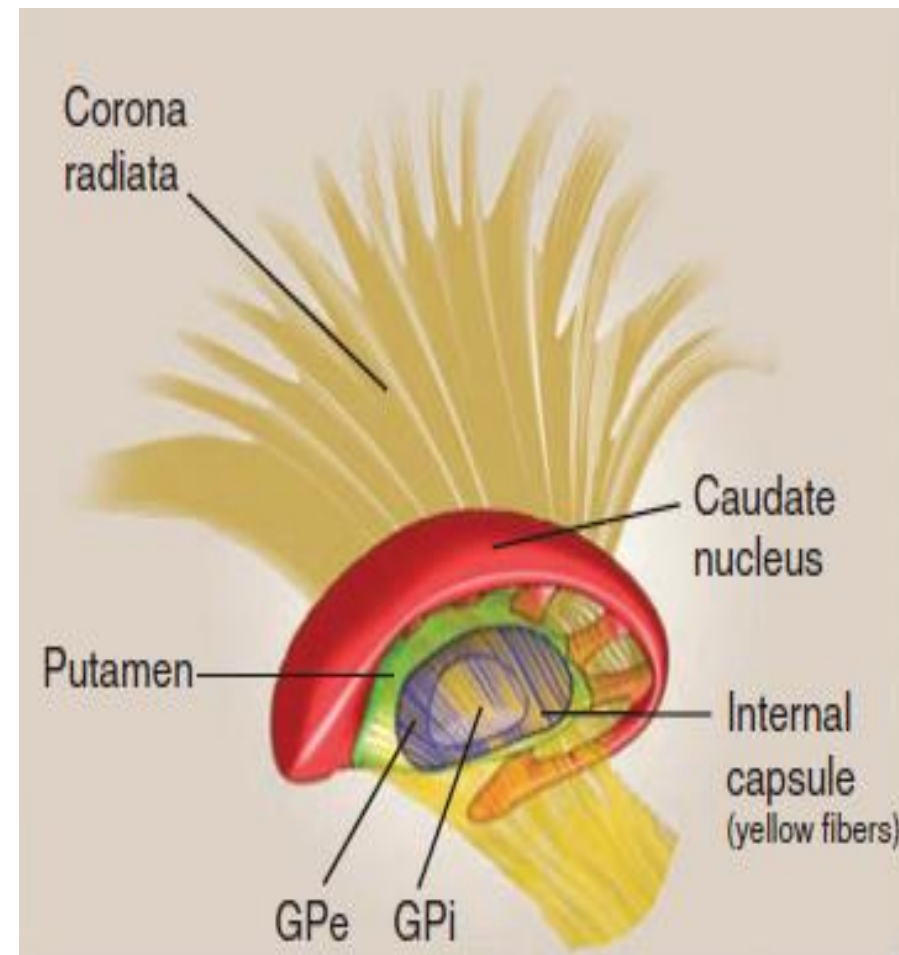


Caudate Nucleus

- Role of caudate circuit
 - Design plans, convert thoughts + ideas into motor actions
 - Determine the **timing and scale of movement**:
 - To what extent the movement will be fast
 - For how long the movement will last
 - Control of eye movement + cognitive + affective behavior

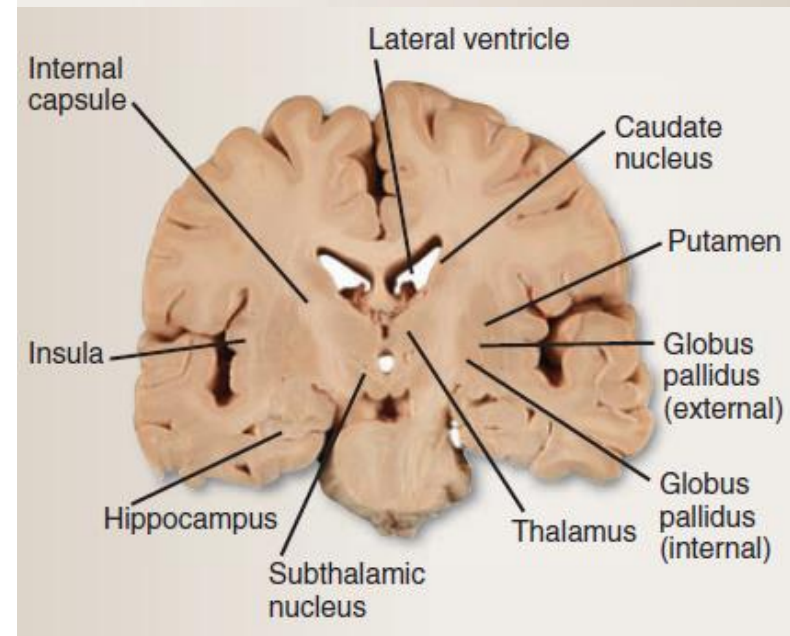
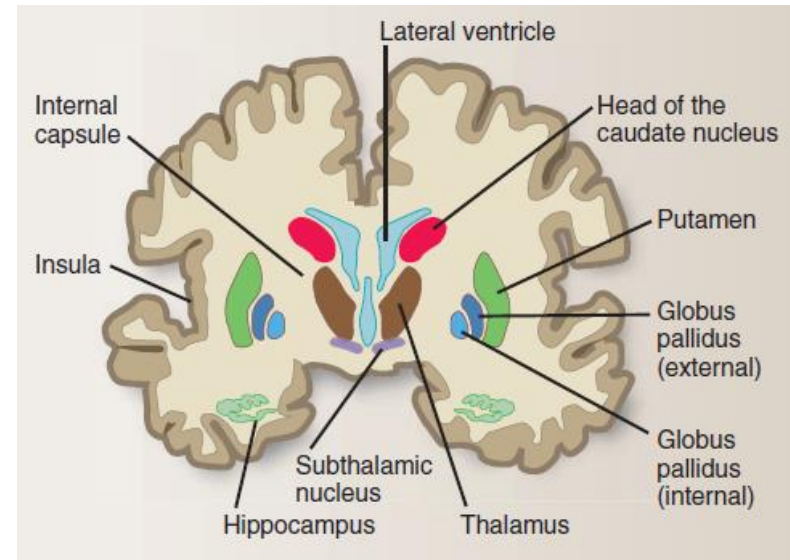
Putamen

- Most lateral of the basal ganglia
- Anterior limb of the internal capsule separates caudate and putamen
- Receive mainly excitatory input from cortical + subcortical structures (+++ thalamus + brainstem)/**motor control**
- *Storage of motor programs of familiar motor actions*



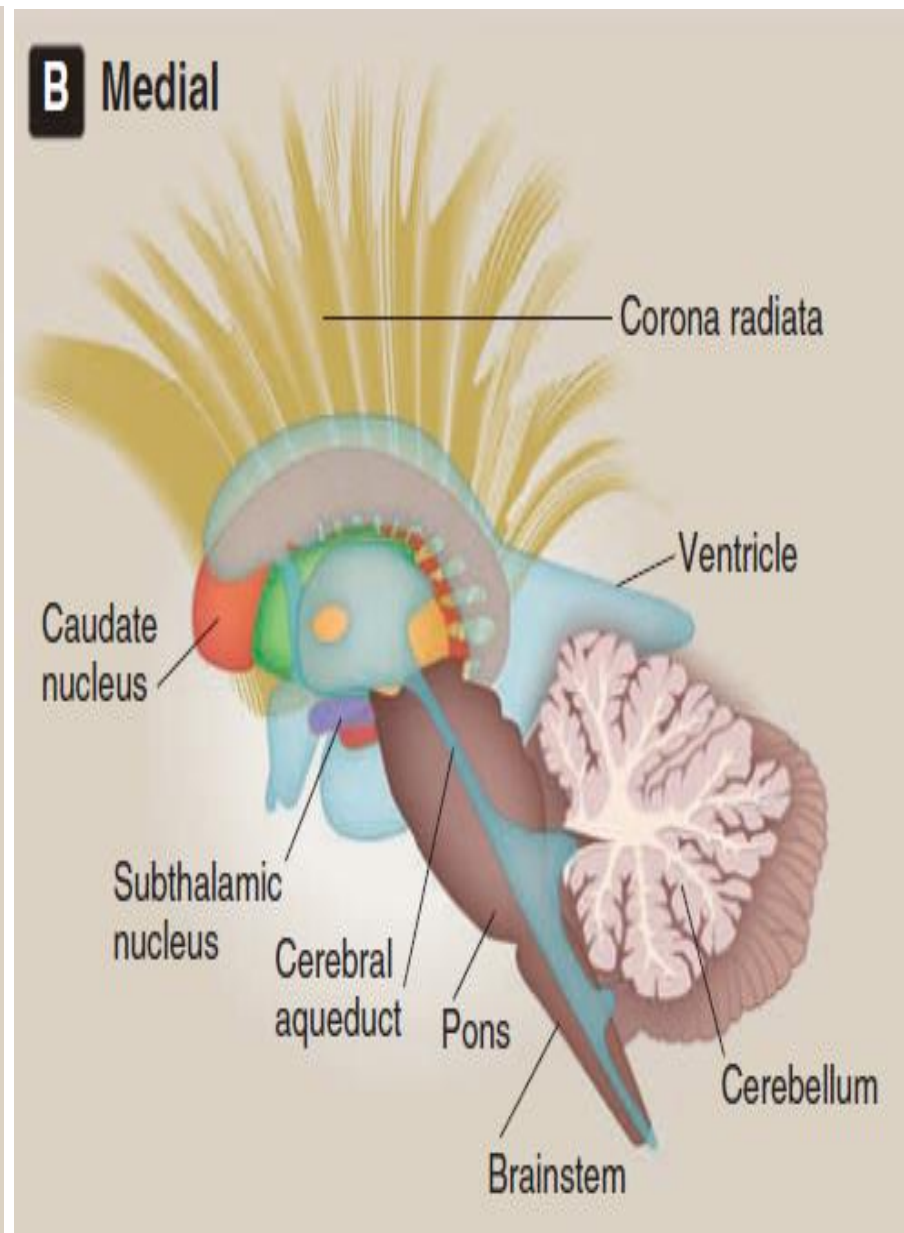
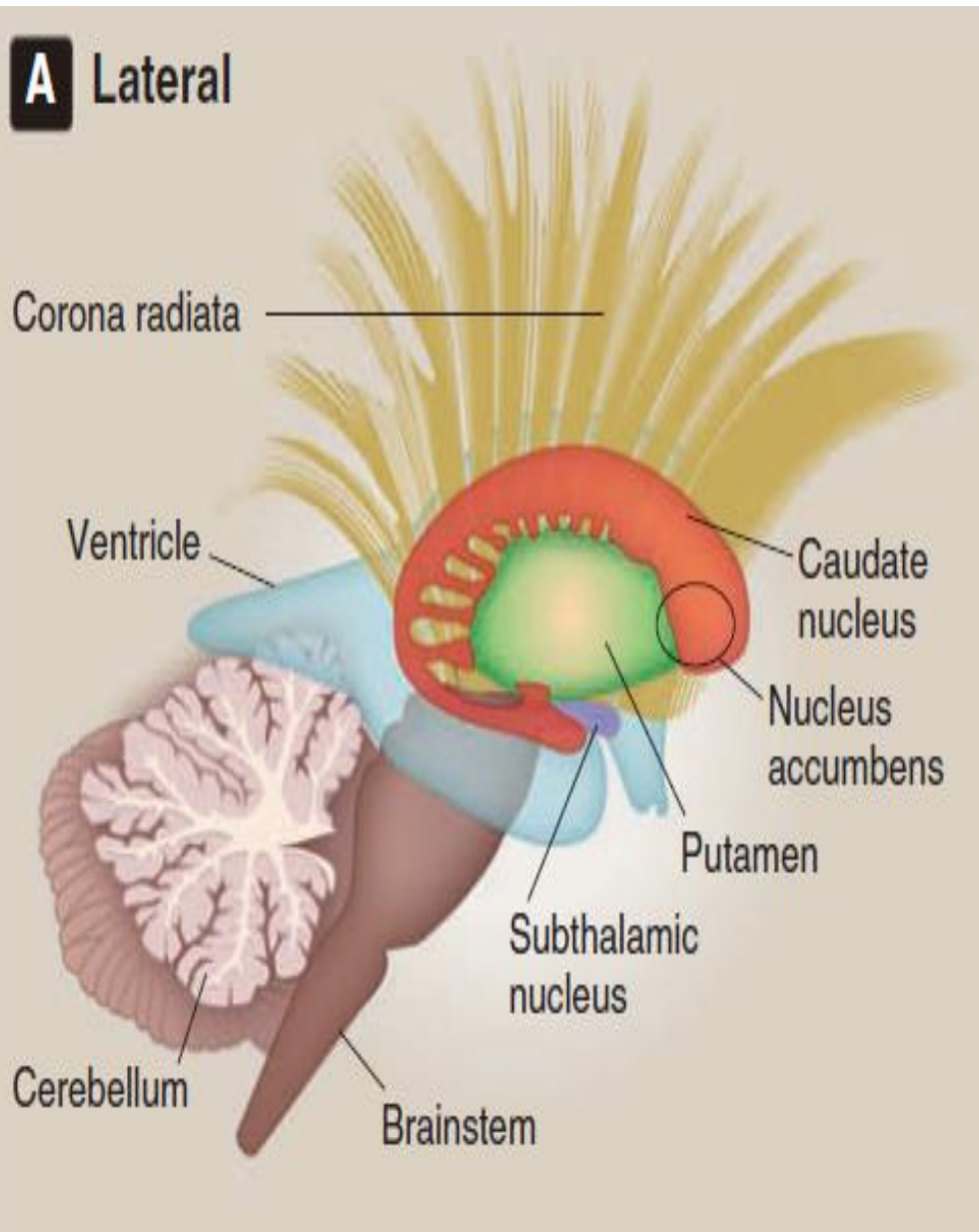
Globus Pallidus/GP

- GP is medial to the putamen + lateral to the thalamus (**GPe + GPi**)
- Internal capsule separates the lentiform nucleus from the caudate nucleus + the thalamus
- GPe + GPi → function + connections are different within the BG
- **GPi** is the **output nucleus** (inhibitory projections to the thalamus)



Nucleus Accumbens

- The anterior + ventral part of the striatum where the **head of the caudate and putamen** are continuous with each other
- Receive extensive **Dopaminergic inputs** from the brainstem and other inputs from the amygdala and parts of the hippocampal formation
- Projects its axons to the substantia nigra and ventral tegmental area
- Integral part of the **limbic** and **reward circuitry**
- Integrate the sequencing of motor responses associated with affective processes
 - **Involved in emotional reinforcement and addiction**

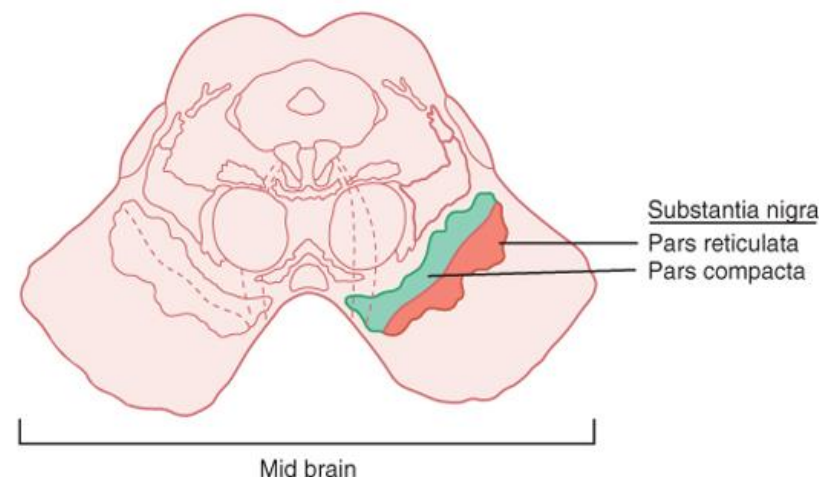
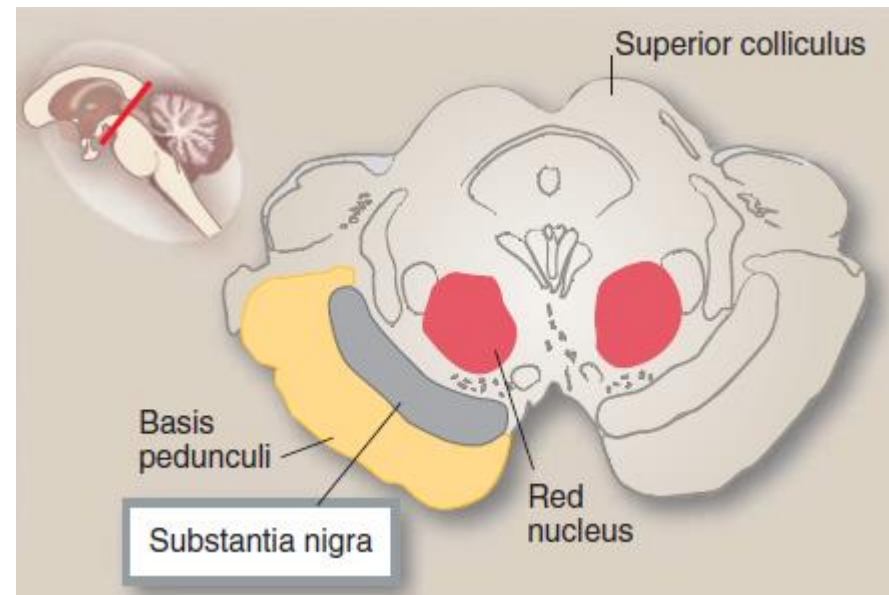


Subthalamic Nucleus/STN

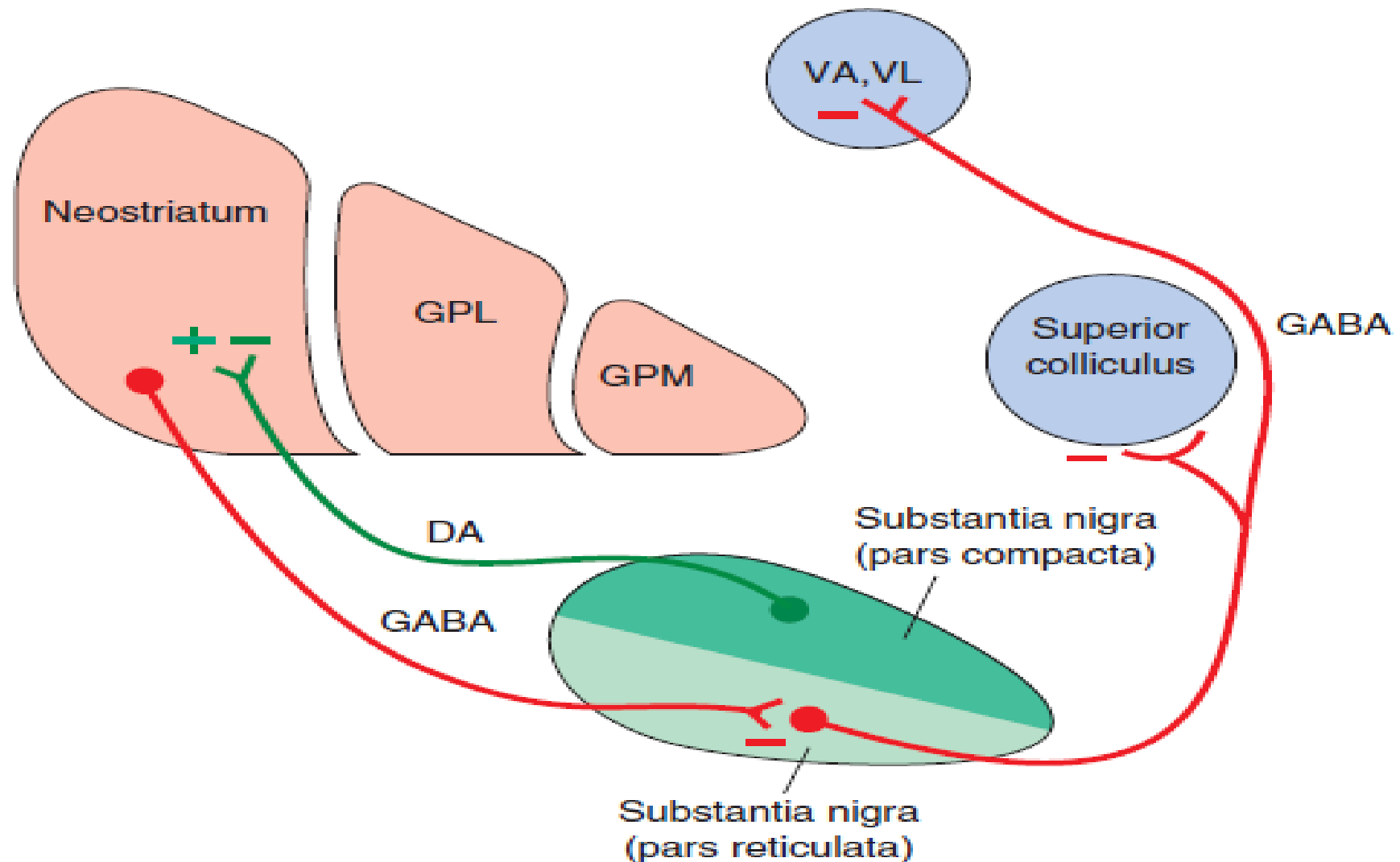
- A biconvex nucleus that lies inferior to the thalamus + superior to the tegmentum of the midbrain + caudal to the hypothalamus
- Receive afferents both from the cortex + other basal ganglia structures
- Output is excitatory thru **glutamatergic projections** to the GP + SN
- Reciprocal connection with the Substantia nigra:
 - Receive **inhibitory signal from the SNpc** and
 - Provide **tonic stimulation to the SNpc** and influence its output
 - Thus, STN **controls the input to the BG and output to the thalamus**

Substantia Nigra

- Located in the rostral midbrain within the cerebral peduncle at the level of the superior colliculus
- SNpc + SNpr
- **Dopaminergic projections (SNpc)** to the putamen + caudate + STN
- **GABAergic projections (SNpr)** to the thalamus + superior colliculus...



Substantia nigra and its connections



Basal Ganglia Circuitry

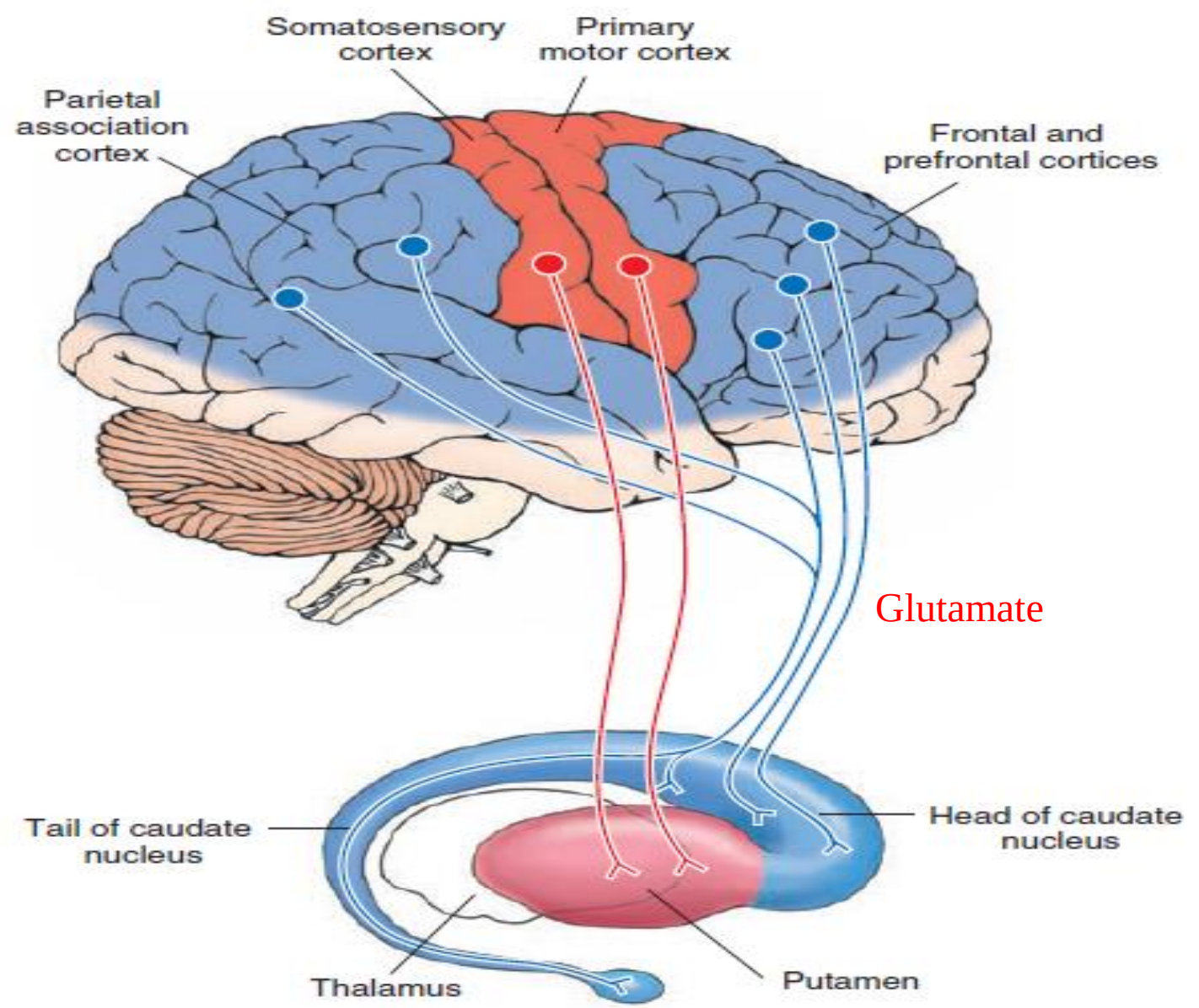
- Inputs

- Inputs terminate to the neostriatum (caudate + putamen)
- A single cell in the striatum receives input from multiple sources (every cell in the striatum is an integrator)
- Caudate + putamen each receive input from distinct cortical + subcortical regions... Intralaminar nuclei of thalamus (CM, PF)... dopaminergic + serotonergic inputs from the midbrain

- ***NB***

- Destruction of Dopaminergic striatum inputs from the substantia nigra are the cause of Parkinson's disease

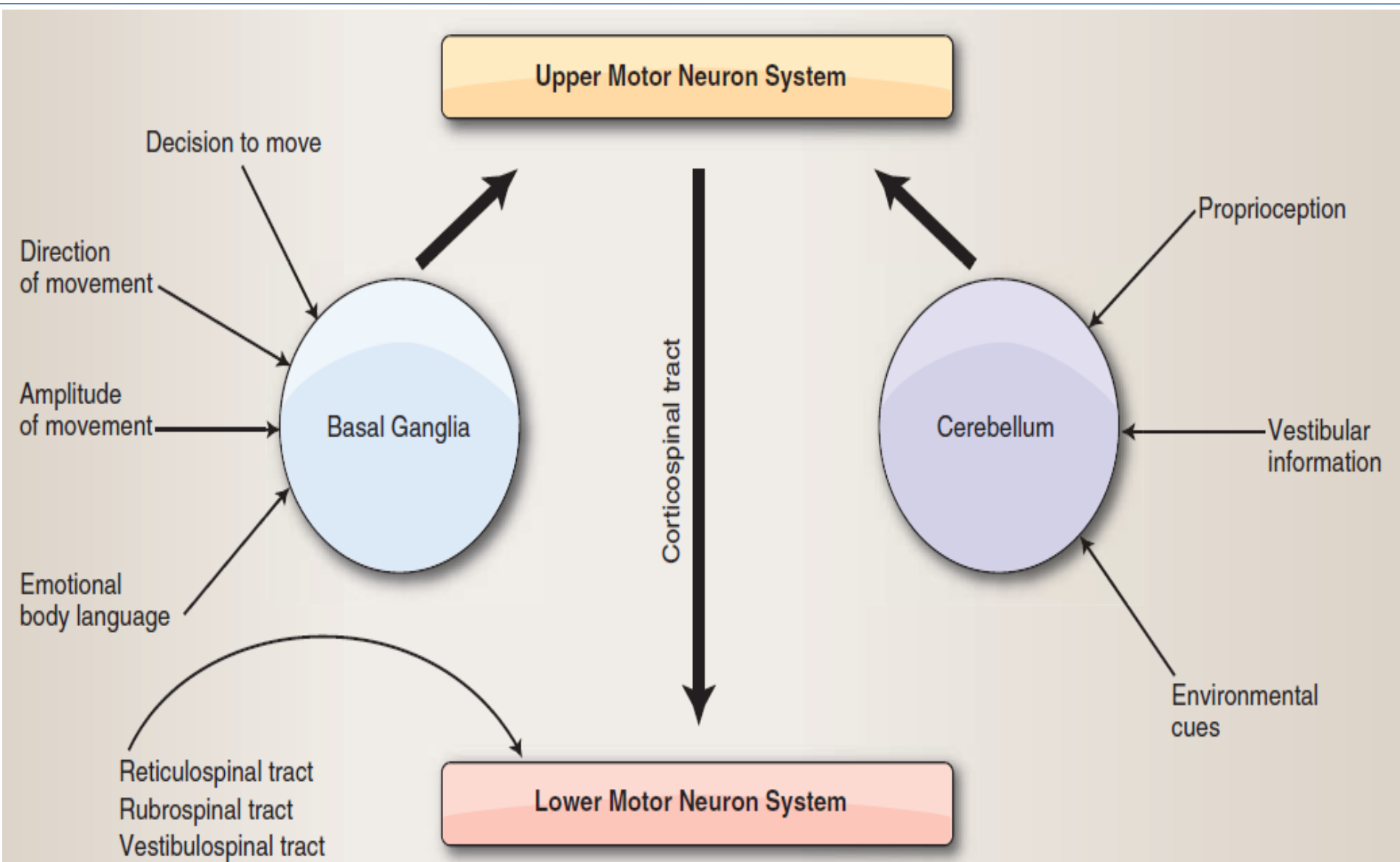
Basal Ganglia Circuitry



Basal Ganglia Circuitry

- Inputs

- Input to the striatum enables the basal ganglia to integrate information from different cortical + subcortical areas
- $\therefore$ Basal ganglia encodes for:
 - The decision to move
 - The direction of movement
 - The amplitude of the movement and
 - The motor expression of emotions



Roles of the basal ganglia in the motor control

Basal Ganglia Circuitry

- **Outputs**
 - BG send major efferents to the cortex via the thalamus
 - Striatum projects to the output nuclei (GPi + SNpr) via direct and indirect pathways
 - Outputs arise from GPi + SNpr → Thalamus (VA + VL) → cortex
 - Output from the BG is inhibitory via **GABAergic neurons**

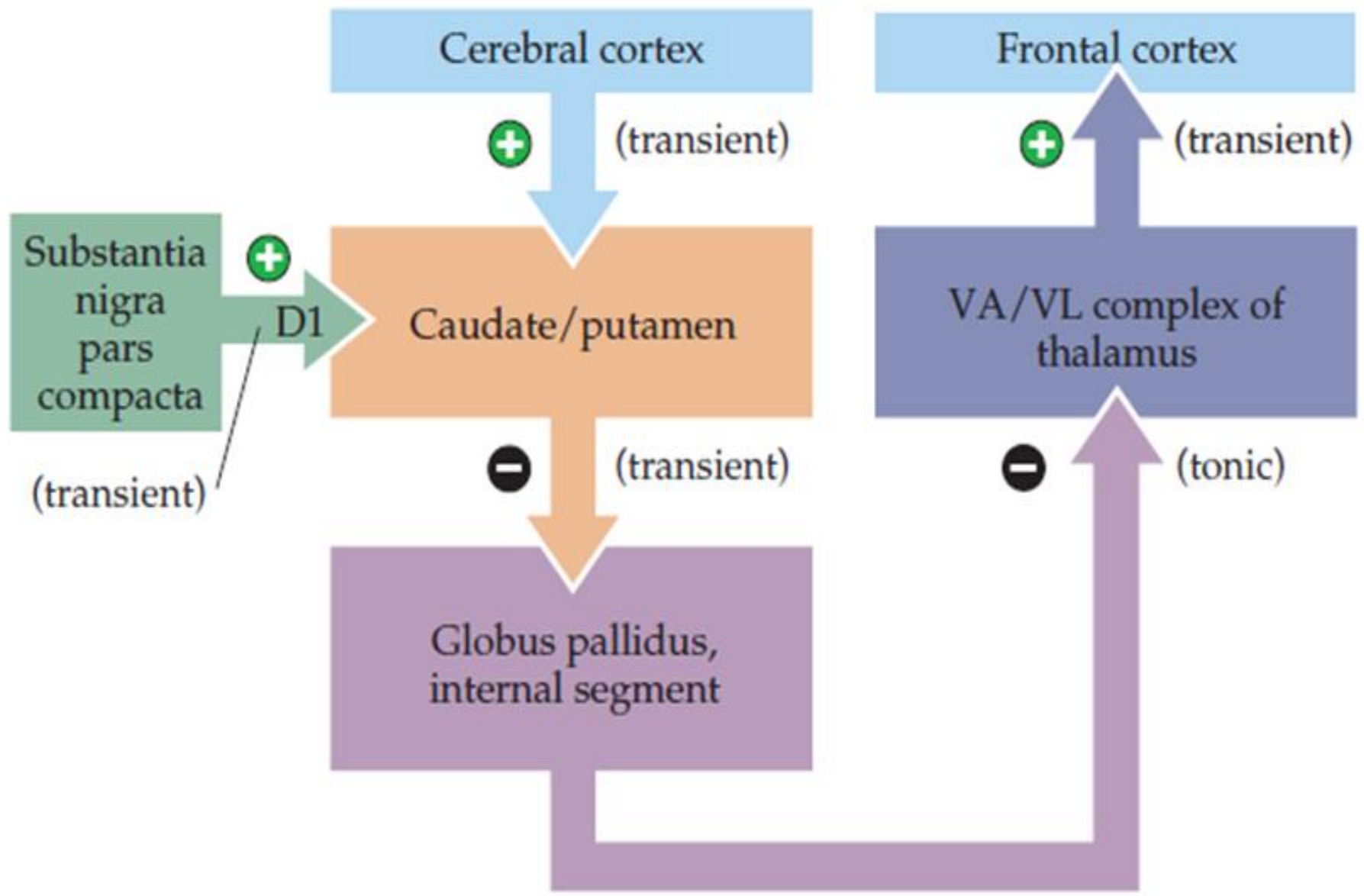
Basal Ganglia Circuitry

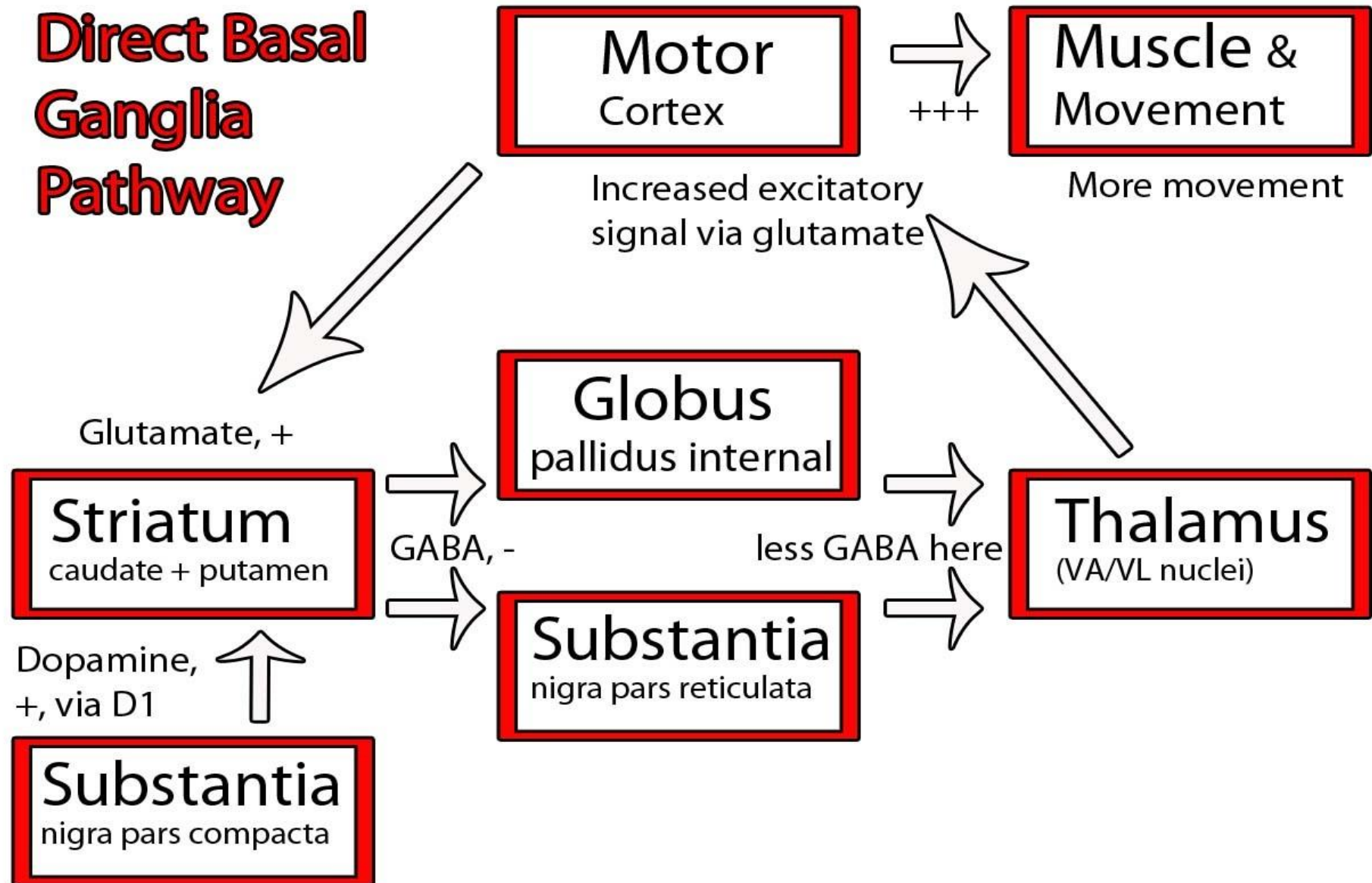
- Internal circuits
 - Input is well integrated + the output is highly regulated
 - Input → Striatum → GPi + SNpr → Output
 - Balance of inhibitory + excitatory pathways → thalamus → cortex
 - Normally, thalamus is under tonic inhibition by the BG
 - Output from the basal ganglia can either decrease or increase the tonic inhibition of the thalamus via two internal pathways:
 - Direct pathway
 - Indirect pathway

Basal Ganglia Circuitry

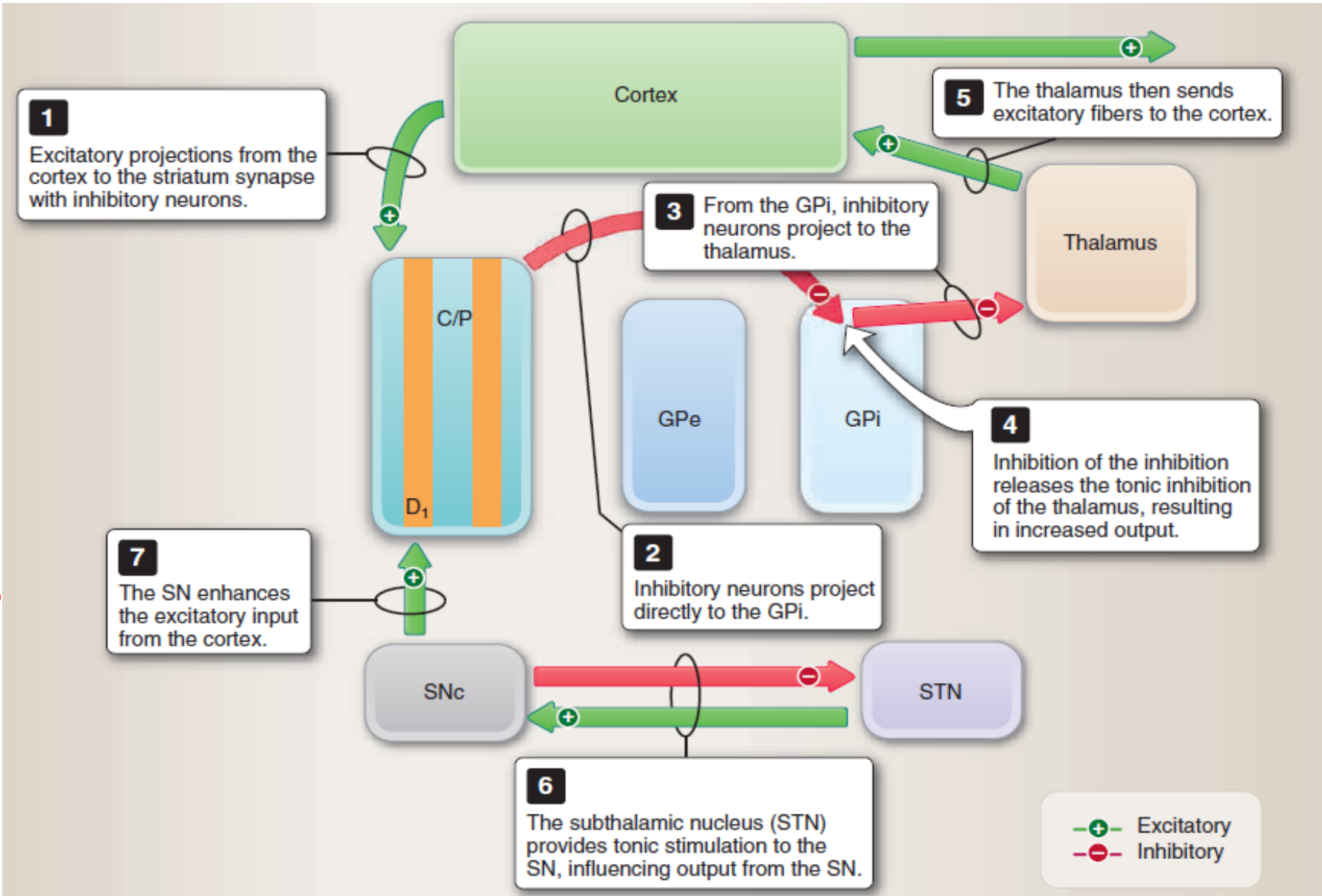
- Direct Pathway
 - Facilitate the streaming of impulses through the thalamus
 - Cortex (corticofugal fibers, Glu/+) → Striatum (-) → GPi + SNpr (-) → Thalamus (+) → Cortex
 - Inhibitory fibers (GPi + SNpr) normally discharge spontaneously and tonically inhibit thalamic discharge to the cortex
 - ∴ Discharge of the direct pathway disinhibits thalamus from tonic inhibition → ↑thalamocortical activity → cortical excitation → more cortical output → facilitates movement (positive feedback)

Direct Pathway





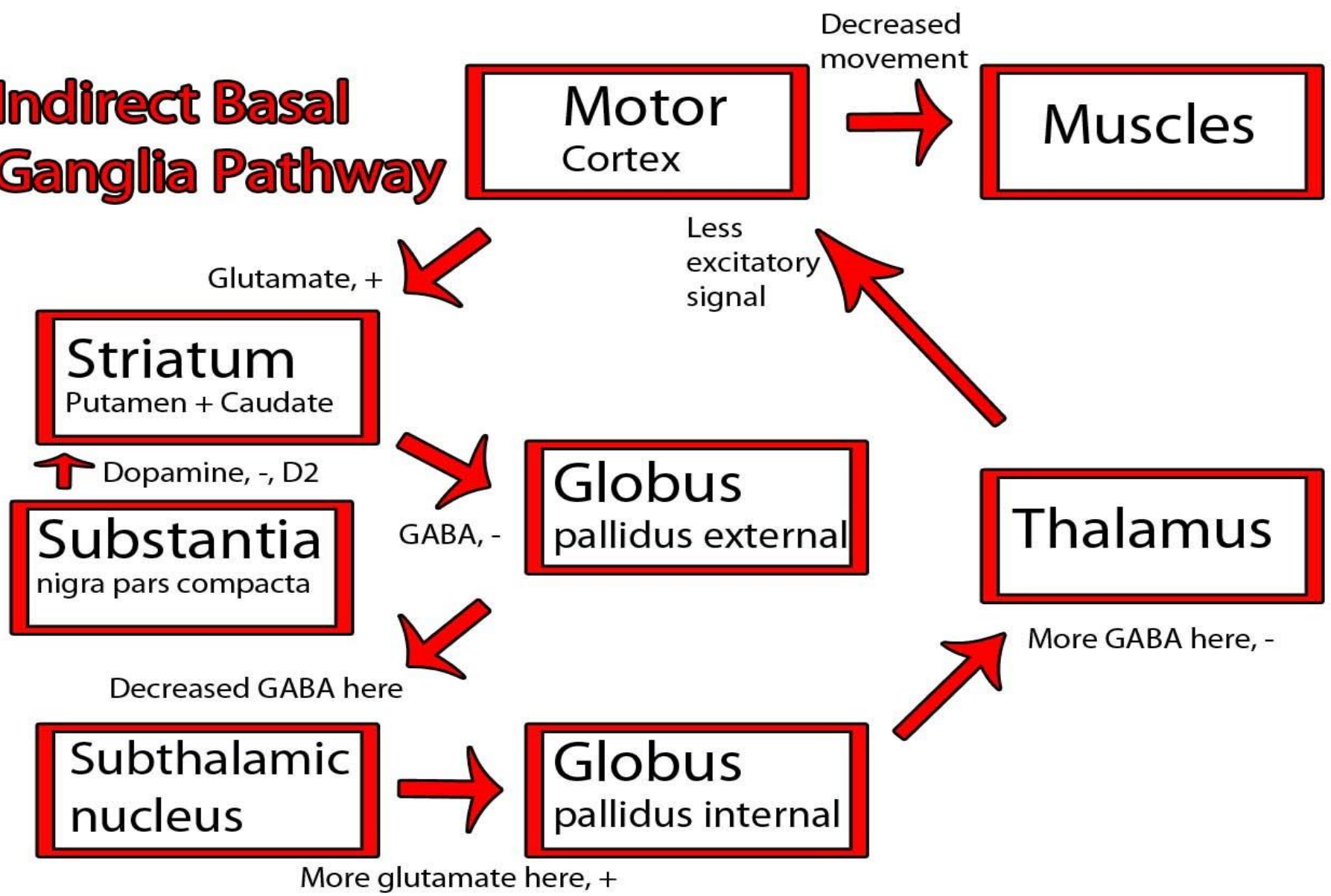
Direct Pathway



Basal Ganglia Circuitry

- Indirect Pathway
 - Inhibits thalamic discharge through an inhibitory Striatopallidal pathway that projects to the GPe
 - GPe neurons fire spontaneously and tonically inhibit the STN
 - However, striatum receives an excitatory projection from the cortex and disinhibits the STN (via pallidum):
 - Cortex (+) → Striatum (-) → GPe (-) → STN (+) → GPi + SNpr (-) → Thalamus (+) → Cortex
- **NB:** STN also receives an excitatory projection from the cortex

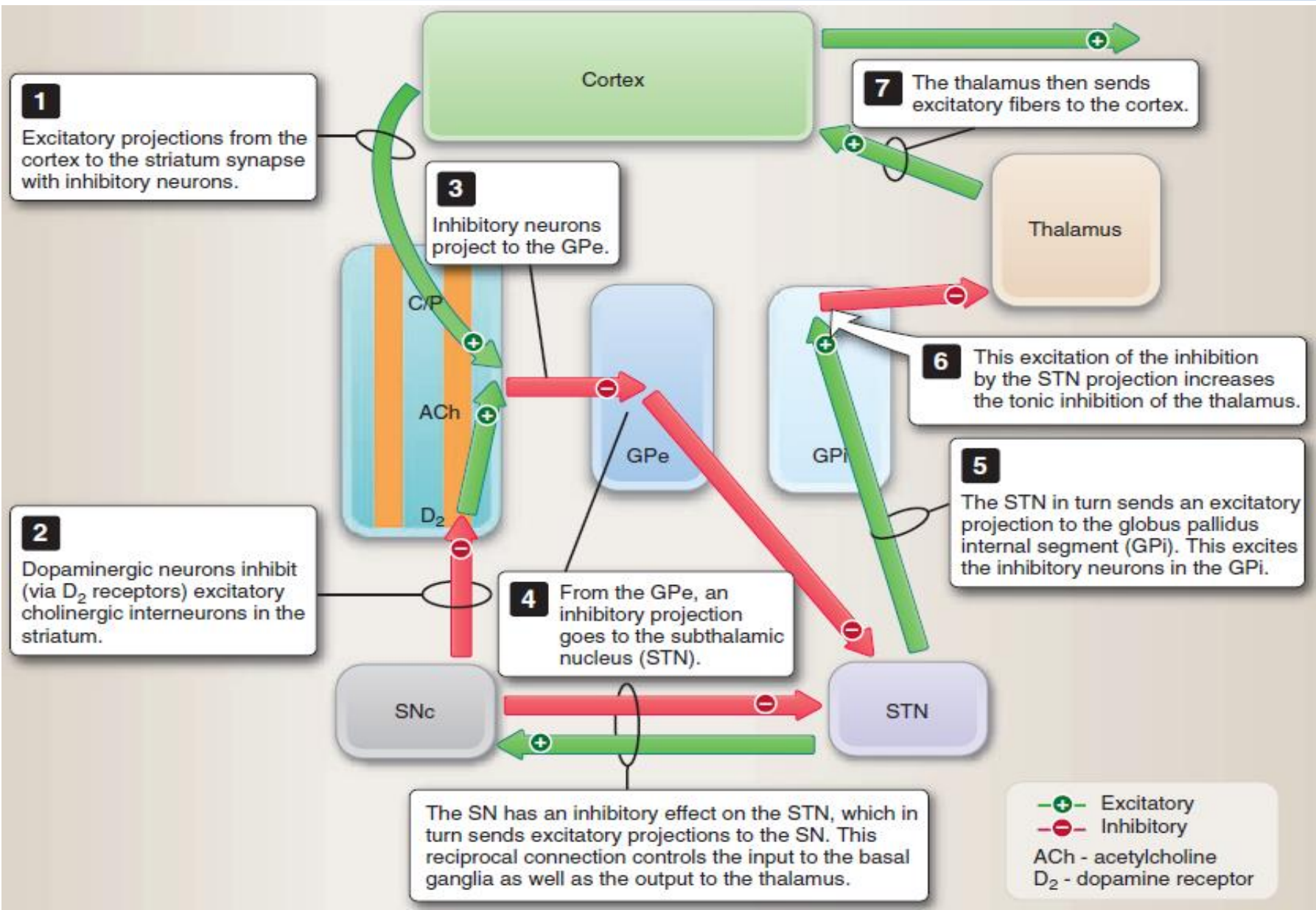
Indirect Basal Ganglia Pathway



Basal Ganglia Circuitry

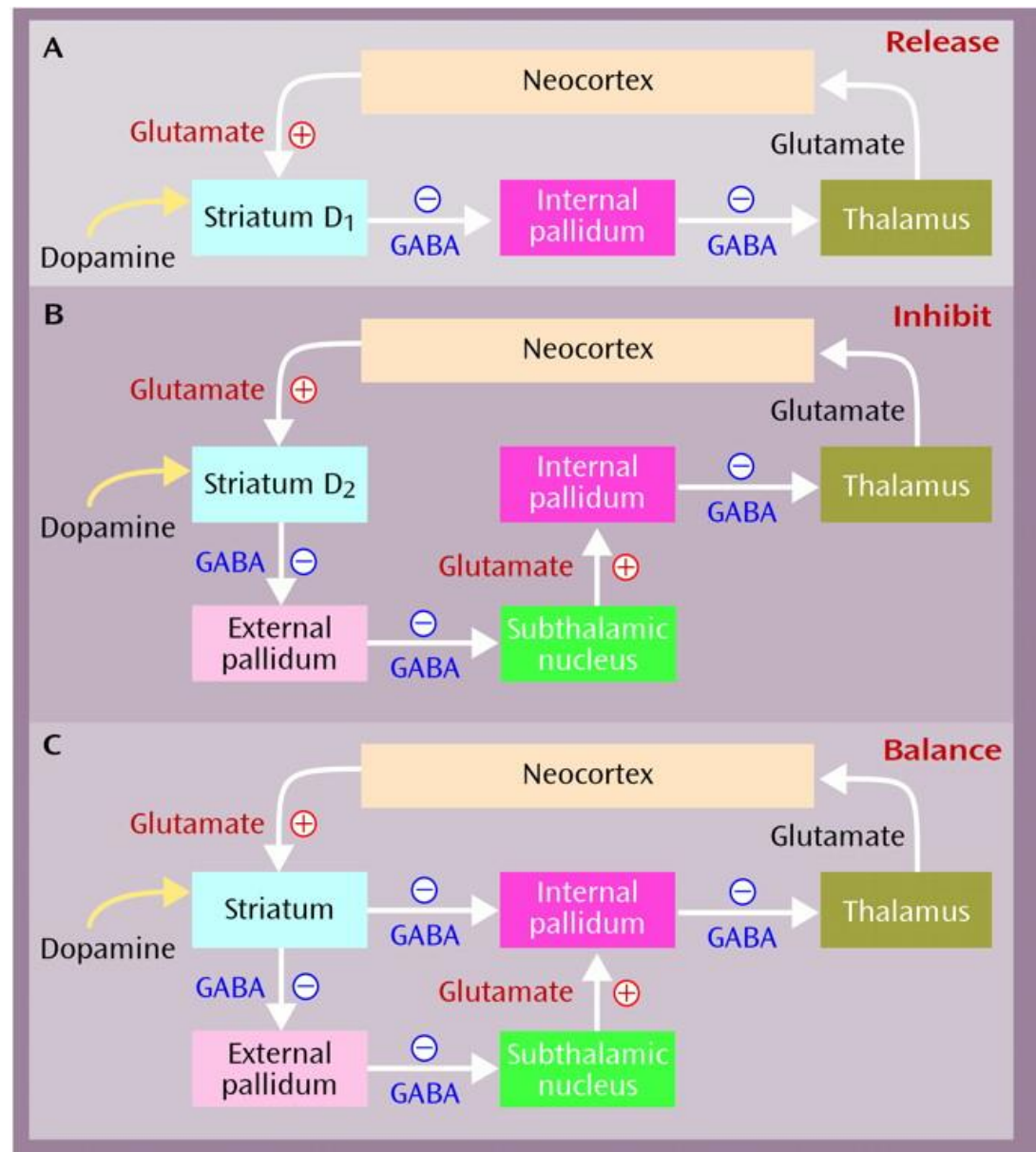
- Indirect Pathway
 - The net result is inhibition of thalamocortical pathway → ↓excitation of the motor cortex → less motor output → inhibits movement (negative feedback):
 - Suppresses superfluous behaviors that are not related to the targeted behavior

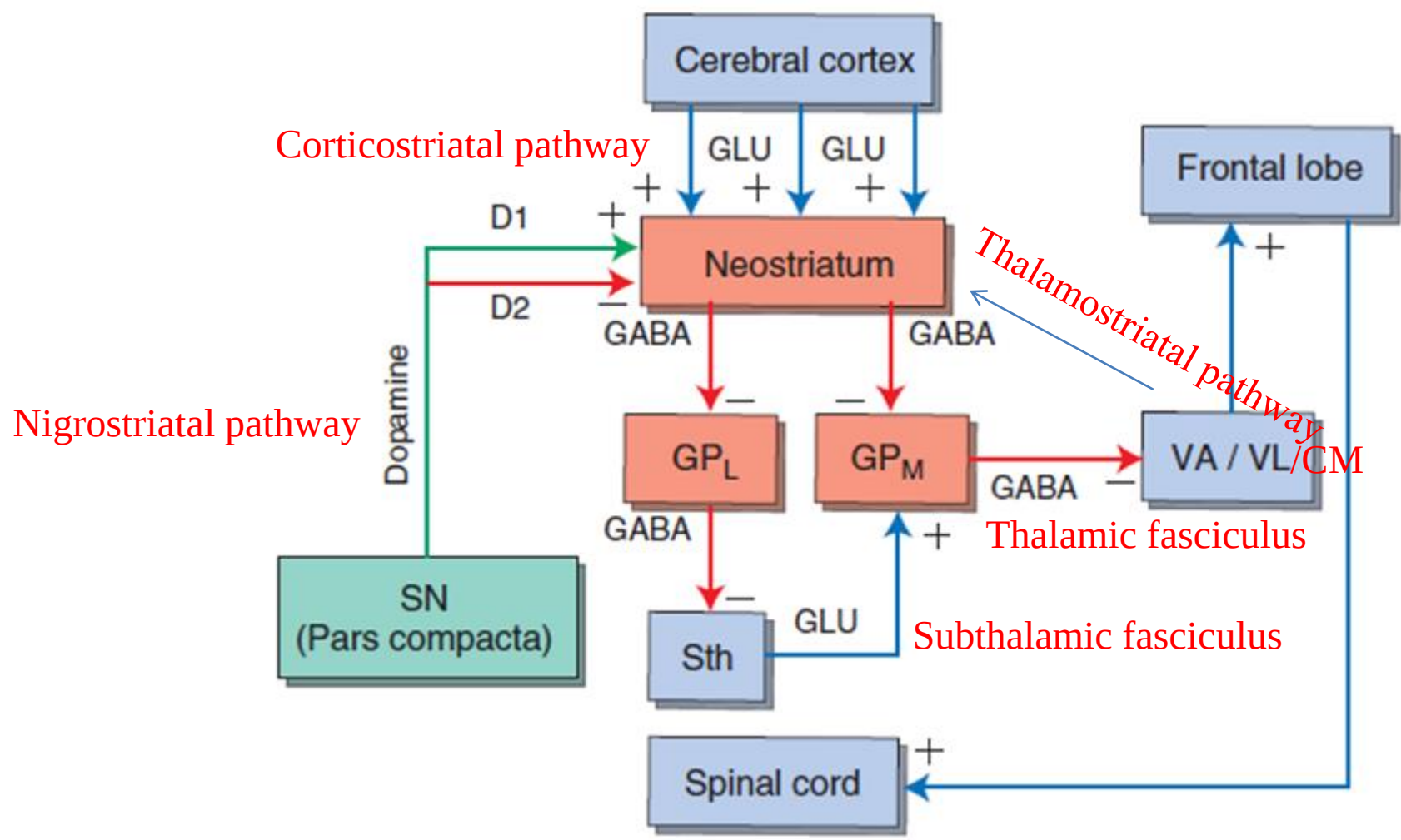
Indirect Pathway



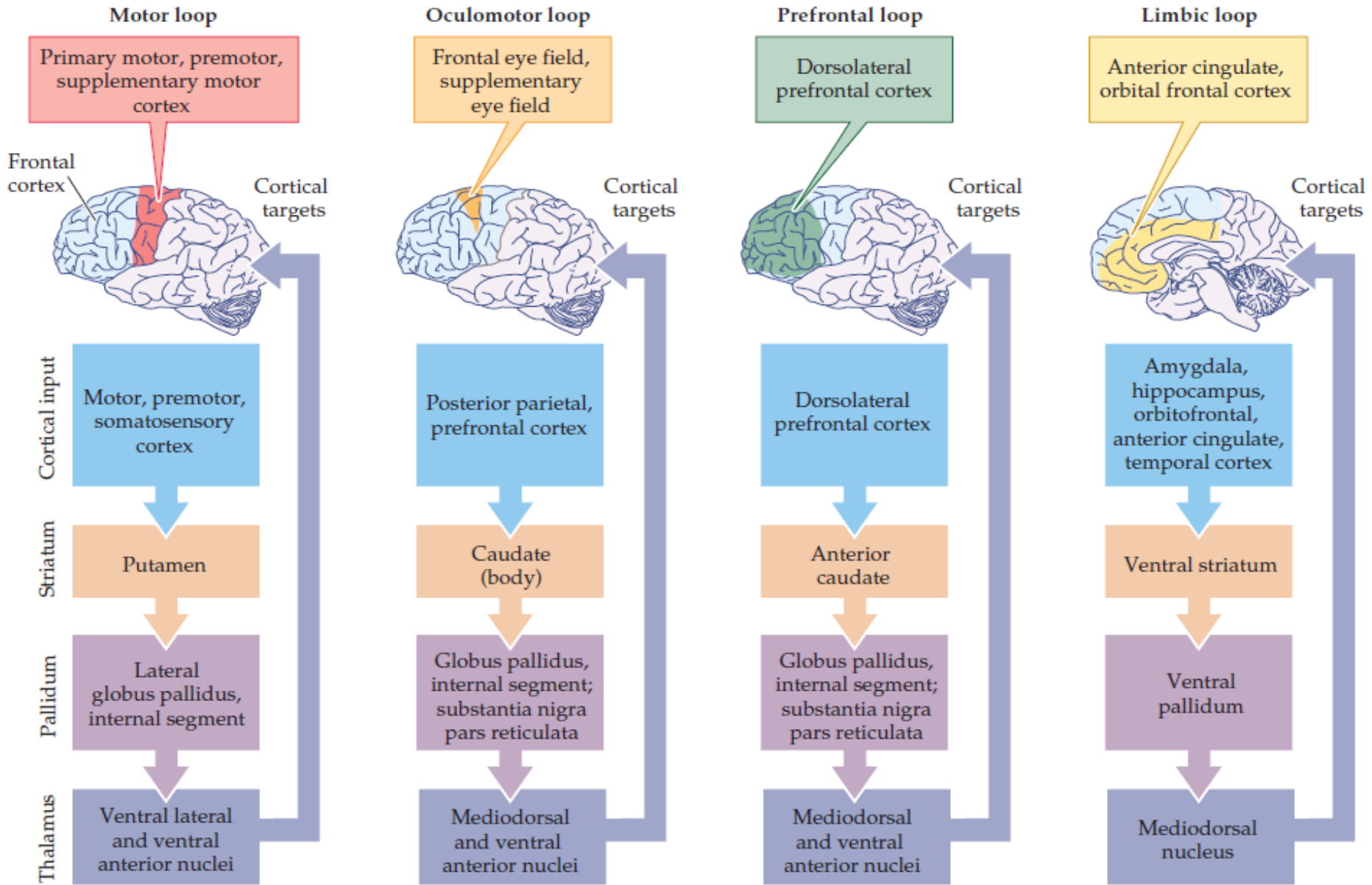
Direct + Indirect Pathways

- Reducing inhibition of thalamocortical neurons → facilitating movements initiated in the cortex





Normal input-output relationships of the basal ganglia



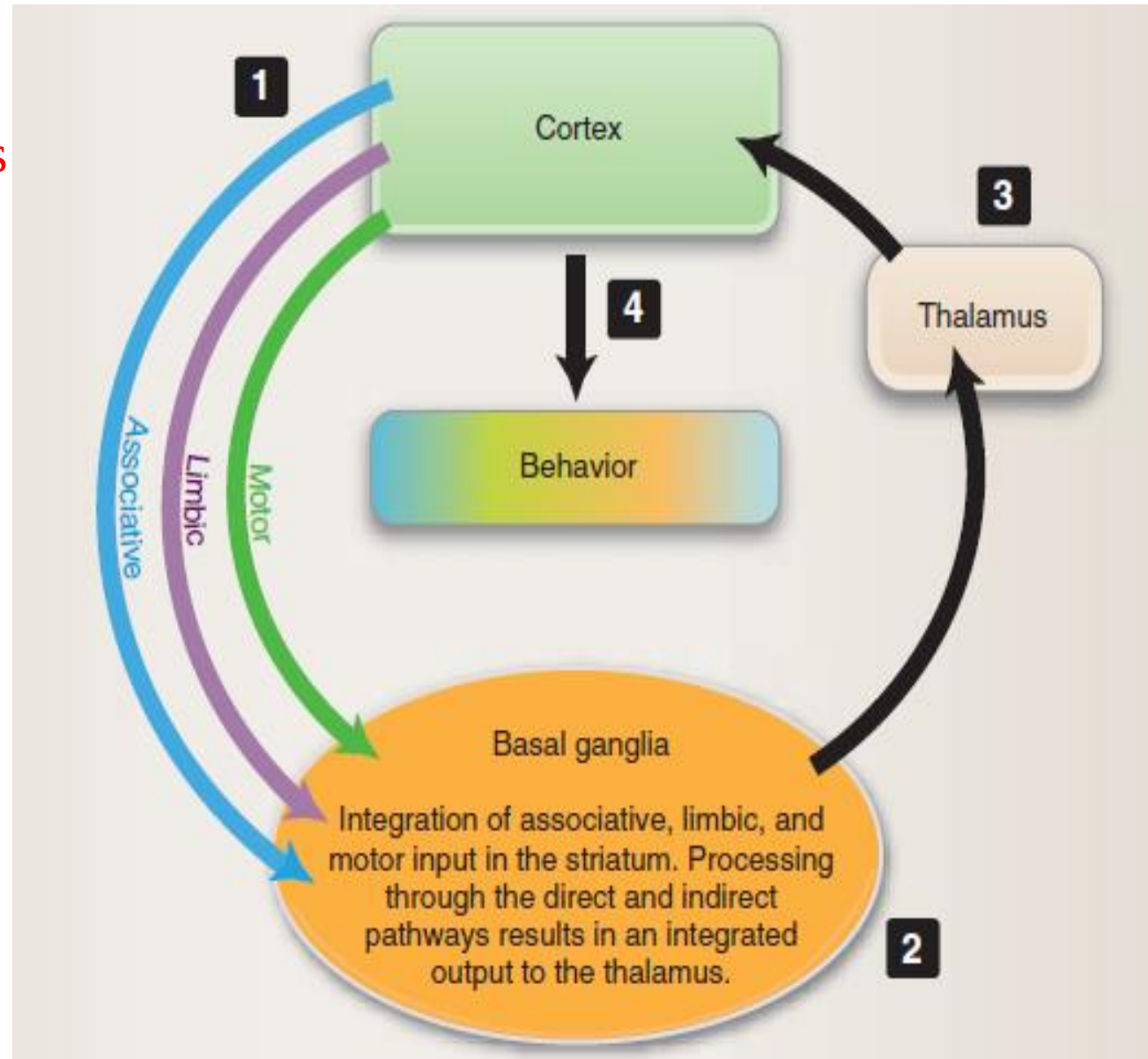
Basal ganglia

Functional relationships

Inputs: Motor

Associative

Limbic



Parkinson's Disease

- Characteristics:
 - Hypokinetic movement disorder
 - Impaired initiation of movement
 - Bradykinesia
 - Increased muscle tone
 - Characterized by loss of nigrostriatal dopamine inputs (degeneration of nigral dopaminergic neurons)
 - Nigrostriatal dopamine inputs excite the direct pathway (D1) and inhibit the indirect pathway (D2)

Parkinson's Disease

- Motor impairments
 - Tremor
 - Rigidity
 - Akinesia and bradykinesia
 - Postural instability
- Postural manifestations:
 - Postural instability
 - Rigidity
 - Stooped

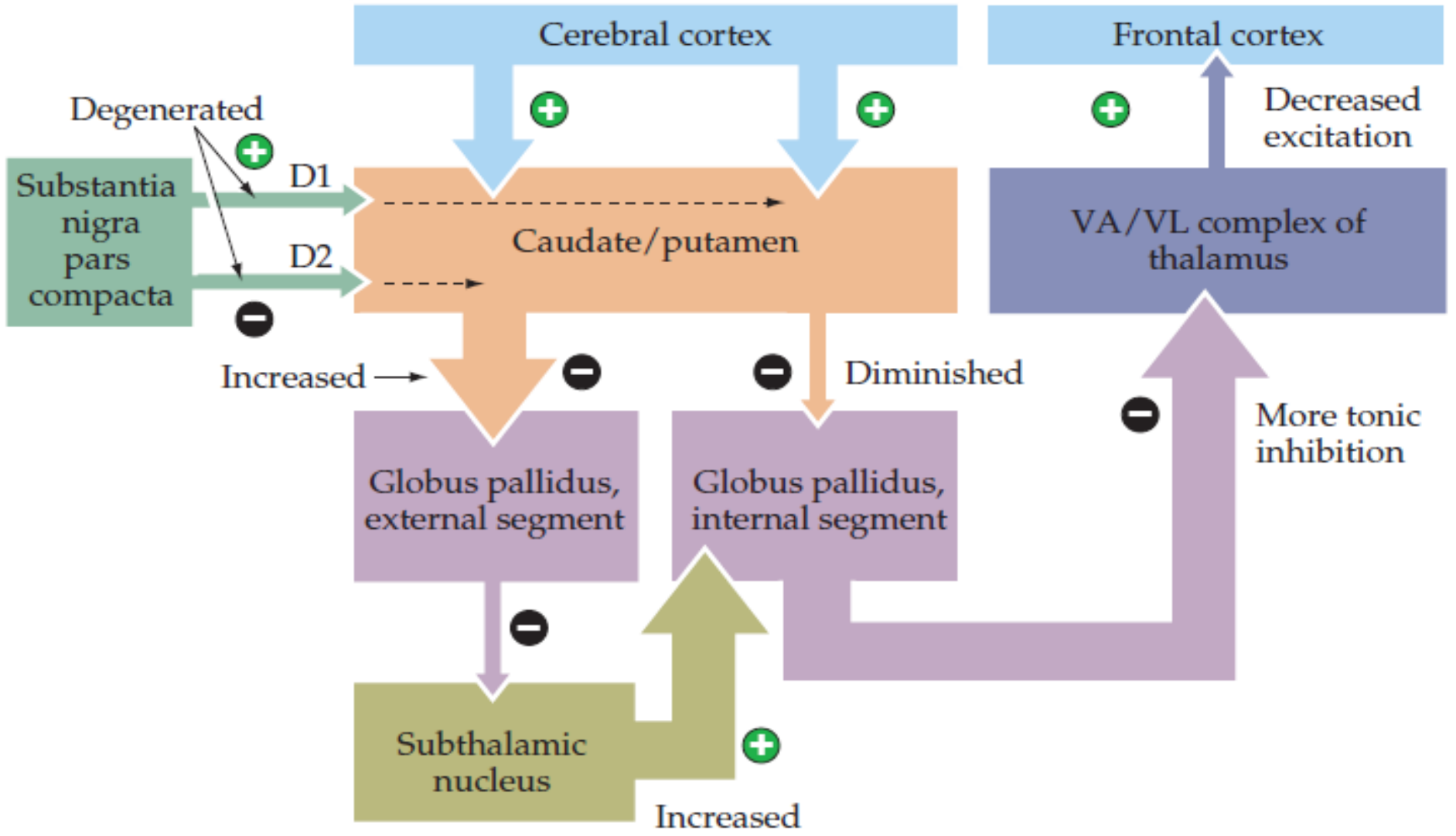
Postural changes cause balance instability

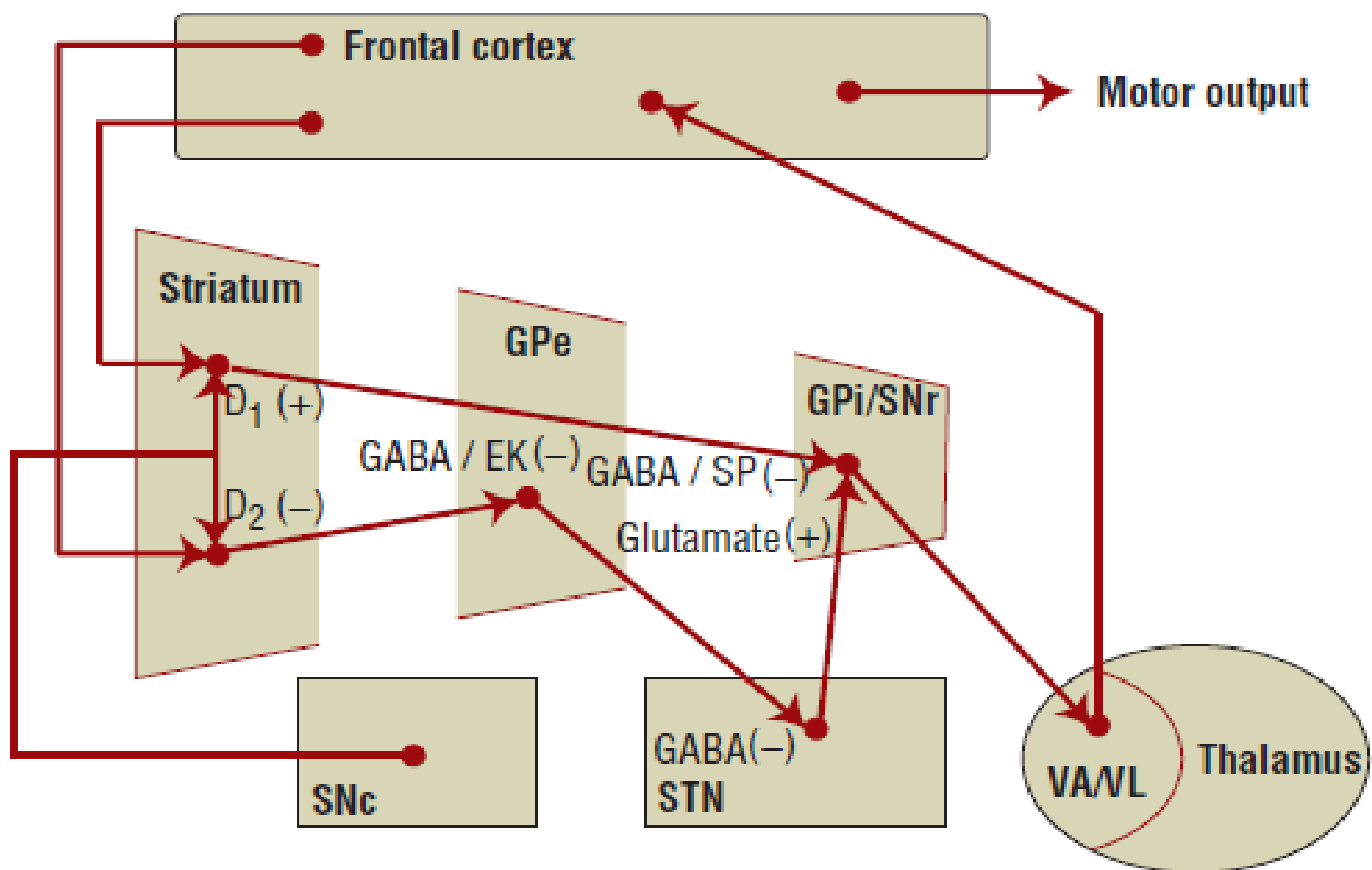


Parkinson's Disease

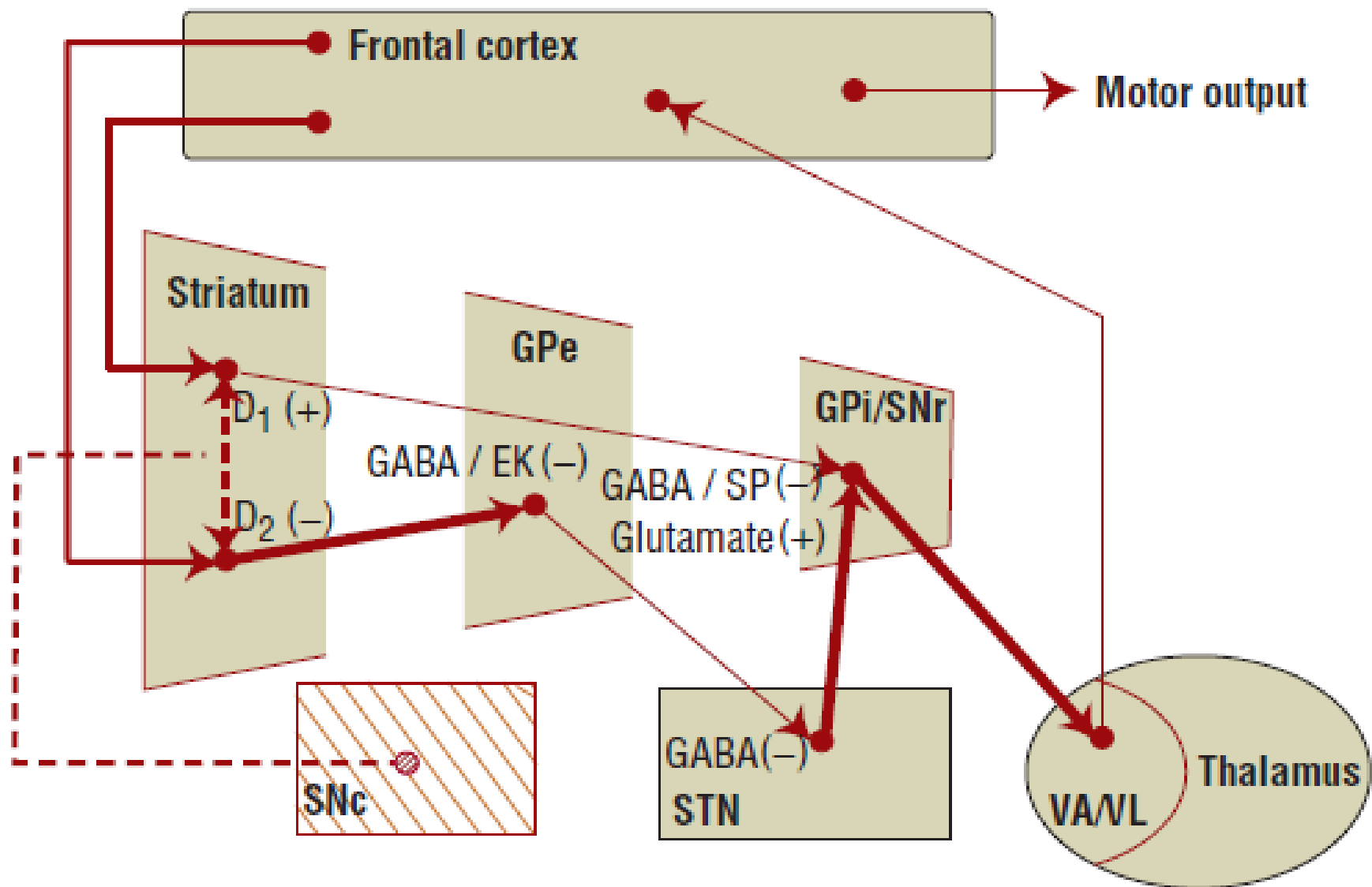
- Characteristics:
 - ↓excitation of the direct pathway (D1) → ↑↑excitation of GPi/SNr → ↑↑inhibition of thalamus → ↑inhibition of motor cortex
(amplified by excitation of the indirect pathway)
 - Loss of dopamine input (D2) → ↑↑excitation of indirect pathway → ↓excitation of GPe → ↓pallidal projections to the STN → ↑↑excitation of GPi/SNr → ↑↑inhibition of thalamus → ↑inhibition of motor cortex

Parkinson's Disease (Hypokinetic)





Normal balance of the basal ganglia–thalamocortical circuit



Nigrostriatal degeneration

Parkinson's Disease

- **NB:**

- Reduced activation of D2 translates into greater inhibition of the thalamus
- Thus, restoring activity at D2 appears to be more important than at D1 for mediating clinical improvements
- Imbalance of dopamine and acetylcholine ($\downarrow$ DA, $\uparrow$ Ach)
- Dopaminergic therapies help to restore motor activity

Huntington's Disease

- Characteristics:
 - Hyperkinetic movement disorder:
 - Marked involuntary movements + decreased muscle tone
 - Preferential loss of GABAergic neurons that project to the GPe compared with the GABAergic neurons that project elsewhere
 - Loss of striatal GABAergic neurons on the indirect pathway
 - $\downarrow$ inhibitory input to GPe $\rightarrow \uparrow\uparrow$ inhibitory output from GPe $\rightarrow \downarrow$ excitatory output from STN $\rightarrow \downarrow$ excitation of GPi/SNr $\rightarrow \downarrow$ inhibition of thalamus $\rightarrow \downarrow\downarrow$ inhibition of motor cortex ($\uparrow$ excitation) $\rightarrow$ excessive motor activity

Huntington's Disease (Hyperkinetic)

